# PHASE 2 – REQUIREMENT ANALYSIS

## Project Title

**Auto Ticket Classification using Flow Designer**

## 1. Introduction

The Requirement Analysis phase identifies the requirements needed to develop the **Auto Ticket Classification using Flow Designer** project.

The school IT helpdesk receives support requests from students and teachers for issues such as **network connectivity, hardware failures, account access problems, and system performance issues**. Currently, IT staff manually review each ticket, identify the issue type, select the appropriate category and subcategory, and inform the caller about ticket creation.

The purpose of this phase is to define what the proposed system must do and what information it needs to store.

## 2. Existing System

In the existing process, IT staff manually handle incoming support requests.

The basic process is:

1. A student or teacher submits an IT support request.
2. IT staff reviews the ticket.
3. The issue type is identified.
4. The appropriate category is selected.
5. The appropriate subcategory is selected.
6. The caller is informed about the ticket creation.

This manual process is time-consuming, can be error-prone, and becomes difficult to manage as the number of requests increases.

# 3. Proposed System

The proposed system uses **ServiceNow Flow Designer** to automatically classify IT tickets.

When a new ticket is created, the system checks the issue description for predefined keywords. Based on the identified issue, the system automatically assigns the appropriate **Category** and **Subcategory**.

The proposed system also sends an automated email notification to the caller after ticket creation.

# 4. Business Requirements

The system must satisfy the following business requirements:

1. Automatically classify IT tickets based on the issue description.
2. Assign both **Category** and **Subcategory** without manual intervention.
3. Support dependent choice logic between Category and Subcategory.
4. Send an automated email notification to the caller upon ticket creation.
5. Store ticket information in a structured and standardized format.
6. Ensure easy maintenance and future scalability.

# 5. Functional Requirements

The system should provide the following functions:

### 5.1 Ticket Creation

The system must allow a user to create an IT support ticket containing the required ticket information.

### 5.2 Automatic Classification

The system must analyze the ticket's short description and identify predefined keywords.

### 5.3 Category Assignment

The system must automatically assign a suitable category based on the identified issue.

### 5.4 Subcategory Assignment

The system must automatically assign the corresponding subcategory.

### 5.5 Dependent Choice

The Subcategory field must depend on the selected Category.

The project defines the following relationships:

| Category | Subcategory |
|---|---|
| Network | Wi-Fi |
| Hardware | Projector |
| Access | Forgot Password |
| Performance | Slow Computer |

These dependent choices are specified in the project document.

### 5.6 Email Notification

After ticket creation and classification, an email notification must be sent to the caller.

## 6. Ticket Data Requirements

The project requires the ticket table to contain the following fields:

| Field Label | Type | Reference / Choices |
|---|---|---|
| Number | Auto Number | — |
| Caller | Reference | sys_user |
| Category | Choice | Network, Hardware, Access, Performance |
| Subcategory | Choice | Wi-Fi, Projector, Forgot Password, Slow Computer |
| Short Description | String | — |

| | | |
|---|---|---|
| Description | String | — |
| State | Choice | New, In progress, On hold, Resolved, Closed |
| Assigned Group | Reference | sys_user_group |
| Assigned to | Reference | sys_user |

These fields and their types are specified in the supplied project document.

## 7. Ticket Classification Requirements

The system should classify common IT issues according to the following logic:

| Keyword / Issue | Category | Subcategory |
|---|---|---|
| WiFi / Network | Network | Wi-Fi |
| Projector | Hardware | Projector |
| Forgot Password | Access | Forgot Password |
| Slow Computer | Performance | Slow Computer |

The Flow Designer conditions described in the project document use the short description to perform these classifications.

## 8. Non-Functional Requirements

The project document identifies the solution as a **no-code, easily maintainable and scalable solution**.

Based on that project requirement, the system should support:

- **Maintainability** – The automation should be easy to maintain.
- **Scalability** – The solution should be capable of being extended in the future.
- **Consistency** – Tickets should be classified using a structured process.
- **Data Accuracy** – Category and Subcategory values should be stored correctly.
- **Standardization** – Ticket information should follow a structured format.

## 9. Email Notification Requirement

The system must send an email notification to the caller after the ticket is created.

The specified email configuration is:

**To:** Caller → Email

**Subject:**
Your Request for the issue has been submitted.

**Body:**
A ticket confirmation message.

This requirement is included in the Flow Designer email action described in the project document.

## 10. User Requirements

The main users involved in the project are:

### Students and Teachers

They create IT support tickets and provide information about their issues.

### IT Helpdesk Staff

They handle and monitor the submitted tickets. The automation reduces the need for them to manually classify each ticket.

The supplied project document describes the helpdesk context as support for students and teachers.

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## 11. Scope of the Project

The project scope includes:

- Creating an IT ticket table.
- Creating required ticket fields.

- Creating Category and Subcategory choices.
- Implementing dependent choices.
- Creating a Flow Designer automation.
- Automatically classifying common IT issues.
- Sending an email notification to the caller.
- Testing the automated classification.
- Validating the stored ticket data.
- Preparing the project for demonstration and documentation.

The supplied project document covers these activities through its configuration, automation, testing, and deployment sections.

## 12. Expected Requirements Outcome

After completing the Requirement Analysis phase, the project requirements are clearly identified.

The system should be able to:

**Create Ticket → Analyze Issue → Identify Keyword → Assign Category → Assign Subcategory → Send Email Notification**

This provides the requirements foundation for designing and developing the Auto Ticket Classification solution.

## 13. Conclusion

The Requirement Analysis phase defines the requirements for the **Auto Ticket Classification using Flow Designer** project.

The system is designed to reduce manual classification by automatically assigning Category and Subcategory values based on the ticket description. It also includes dependent choices, structured ticket information, and automated email notification.

These requirements will be used as the basis for the **Project Design and Development phases**.